

MACHINE LEARNING INTERNSHIP

Assignment 1: Statistics, Probability & Python Foundations

Setup Requirement

Ensure the following are ready before starting:

- Python 3.x with Jupyter Notebook (or use Google Colab).
- Required libraries: **pip install numpy pandas matplotlib scipy**

Task 1: Descriptive Statistics, Probability & Combinatorics (Manual)

Solve the following problems with full workings (formulas and step-by-step calculations). Use a calculator if needed but show all intermediate steps.

1.1 Summary Statistics

The marks scored by 10 students in a Mathematics test (out of 100) are:

Data: 72, 85, 90, 65, 78, 92, 68, 75, 88, 80

Calculate the following:

- Mean
- Median
- Mode (state if no mode exists)
- Range
- Variance and Standard Deviation (use the sample formula, divide by n-1)

1.2 Probability

Q1. A fair coin is tossed 3 times. What is the probability of getting **exactly 2 heads**?

Q2. A bag contains 5 red, 4 blue, and 3 green balls. One ball is drawn at random. Find the probability that it is:

- Red
- Not green

Q3. Two dice are rolled together. What is the probability that the sum of the numbers is **greater than 8**?

1.3 Permutations and Combinations

Q1. In how many ways can the letters of the word **MACHINE** be arranged so that all vowels come together?

Q2. A committee of 4 members is to be formed from 6 men and 5 women. In how many ways can it be formed if it must include exactly 2 men and 2 women?

1.4 Discrete Probability Distribution (Binomial)

A coin is biased such that the probability of getting a head in a single toss is 0.6. The coin is tossed 5 times. Using the Binomial distribution formula, find the probability of getting:

- Exactly 3 heads
- At least 4 heads

Formula: $P(X = k) = C(n, k) \times p^k \times (1-p)^{(n-k)}$

Task 2: Python, NumPy & Pandas (Hands-On)

2.1 Verify Statistics with NumPy

Using the same data from Task 1.1, write Python code with NumPy to verify your manual calculations.

```
import numpy as np

marks = np.array([72, 85, 90, 65, 78, 92, 68, 75, 88, 80])

print("Mean:", np.mean(marks))
print("Median:", np.median(marks))
print("Range:", np.max(marks) - np.min(marks))
print("Variance (sample):", np.var(marks, ddof=1))
print("Std Deviation (sample):", np.std(marks, ddof=1))
```

2.2 Pandas DataFrame Operations

Create the following dataset in Python (you may save it as **employees.csv** or build it directly using a dictionary):

emp_id	name	department	age	salary	experience_yrs
E01	Aarav	IT	28	55000	3
E02	Bhavya	HR	32	48000	5
E03	Chirag	IT	26	52000	2
E04	Diya	Finance	35	72000	8
E05	Eshan	IT	30	60000	4
E06	Farah	Sales	29	45000	3
E07	Gaurav	Finance	40	85000	12
E08	Hina	HR	27	42000	2
E09	Ishan	Sales	33	55000	6
E10	Jiya	IT	38	78000	10

Perform the following operations and submit code + screenshots:

- Load the data into a Pandas DataFrame.
- Display the first 5 rows using **df.head()**.
- Print summary statistics using **df.describe()**.
- Find the **average salary by department** using **groupby**.
- Filter and display employees with **experience > 5 years**.
- Sort the DataFrame by salary in **descending order** and display the top 3.
- Add a new column **salary_per_year_exp** = salary / experience_yrs.

Task 3: Inferential Statistics + Visualization

3.1 Hypothesis Testing (One-Sample t-Test)

A company claims that the average daily working hours of its employees is **8 hours**. A random sample of 10 employees gives the following daily working hours:

Data: 7.5, 8.2, 7.8, 9.0, 8.5, 7.2, 8.8, 9.1, 7.4, 8.0

Test the claim at a **5% significance level** (alpha = 0.05) using a one-sample t-test.

- Write the Null and Alternative hypotheses (H0 and H1).

- Calculate the t-statistic manually (formula: $t = (\bar{x} - \mu) / (s / \sqrt{n})$).
- Find the p-value or critical t-value (degrees of freedom = $n - 1 = 9$).
- Verify your result using Python:

```
from scipy import stats
```

```
data = [7.5, 8.2, 7.8, 9.0, 8.5, 7.2, 8.8, 9.1, 7.4, 8.0]
t_stat, p_value = stats.ttest_1samp(data, popmean=8)
print("t-statistic:", t_stat)
print("p-value:", p_value)
```

- State your conclusion: do we reject or fail to reject H_0 ?

3.2 Confidence Interval

Using the same working hours data, calculate the **95% confidence interval** for the population mean. Use the formula: $CI = \bar{x} \pm (t_{critical} \times s / \sqrt{n})$

3.3 Matplotlib Visualization

Using the employees dataset from Task 2, create the following 3 plots with proper titles, axes labels, and colours:

- **Bar Chart:** Department vs Average Salary.
- **Scatter Plot:** Experience (years) vs Salary, with each department in a different colour.
- **Histogram:** Distribution of employee ages (bins of 5).

Deliverables

- Task 1: Hand-written or typed solutions with all workings for summary statistics, 3 probability problems, 2 combinatorics problems, and the binomial distribution problem.
- Task 2: Python code with screenshots of NumPy verification and all Pandas operations on the employees dataset.
- Task 3: Hypothesis test calculation and Python verification, confidence interval calculation, and 3 matplotlib plots with code and screenshots.

How to Submit

1. Create the assignment in MS Word (.doc / .docx) or PDF format.
2. Include your full name, course / internship name, and assignment number at the top of the document.
3. Show all formulas, calculations, code, and screenshots clearly.
4. Upload the completed file to your Google Drive.
5. Enable sharing access by setting the file to "Anyone with the link can view".
6. Copy the Google Drive link and submit it before the assignment deadline through the official submission portal.